DAY 03 • STUDENT ACTIVITIES

The Cyber Threat Landscape and Attack Frameworks

Malicious software behaviors, threat actors, the Cyber Kill Chain and MITRE ATT&CK

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats
Coverage	Required Knowledge 1.10 / 5.1 Malicious software behaviors; 5.2 Threats; 1.11 / 5.3 Knowledge of attack frameworks
Duration	8 hours (1 training day)

Trainee Name	Batch / Section	Date

Activities Overview

No.	Activity	Type	Time	Assets Needed
3.1	Malware Behavior Matching	Individual	40 min	malware_behavior_cards.txt (lab: ~/day3)
3.2	Kill Chain Mapping of a Real-World-Style Incident	Group (3–4)	60 min	Incident_Narrative_Kill_Chain.docx (lab: ~/day3, as .txt)
3.3	Exploring MITRE ATT&CK	Individual (hands-on)	60 min	Internet browser (attack.mitre.org)
3.4	IOC or IOA?	Pair	20 min	None

CAUTION — Lab Rules

- Work only on the lab machines, virtual machines and files provided by your trainer.
- Never scan, probe or test any system outside the authorized training network.
- Treat all sample logs and data as confidential, as if they belonged to a real client.

Activity 3.1 — Malware Behavior Matching

Type	Individual
Time Allotment	40 min
Objective	Identify the malware type from observed behavior.
Assets / Materials	assets/malware_behavior_cards.txt (CTM Virtual Lab: ~/day3/)

Instructions

1. Open assets/malware_behavior_cards.txt (in the CTM Virtual Lab: cat ~/day3/malware_behavior_cards.txt). Each card describes behavior seen by a security tool.
2. Identify the most likely malware type and the ATT&CK tactic for each card.
3. Rate the urgency (Low, Medium, High, Critical).

Work Space

Card	Malware Type	ATT&CK Tactic	Urgency	Key Clue
1				
2				
3				
4				
5				
6				
7				
8				

Deliverable

Completed matching table (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
At least 6 of 8 malware types correctly identified	☐	☐
Tactics are correctly mapped	☐	☐
Urgency ratings are justified	☐	☐

Activity 3.2 — Kill Chain Mapping of a Real-World-Style Incident

Type	Group (3–4)
Time Allotment	60 min
Objective	Map events from an incident narrative to the Cyber Kill Chain.
Assets / Materials	assets/Incident_Narrative_Kill_Chain.docx (CTM Virtual Lab: cat ~/day3/Incident_Narrative_Kill_Chain.txt)

Instructions

1. Read the incident narrative in the asset document.
2. Place each numbered event into the correct Kill Chain phase.
3. Identify the earliest phase where the SOC could have stopped the attack, and how.

Work Space

Kill Chain Phase	Event No(s).	Evidence / Log Source
Reconnaissance		
Weaponization		
Delivery		
Exploitation		
Installation		
Command & Control		
Actions on Objectives		

Earliest point the attack could have been stopped, and how:

Deliverable

Kill Chain map (worksheet in your CTM Virtual Lab notebook), presented to the class.

Performance Criteria (Trainer Use)

Criteria	Yes	No
Events are placed in the correct phases	☐	☐

Criteria	Yes	No
Evidence sources are identified	☐	☐
A realistic early detection point is proposed	☐	☐

Activity 3.3 — Exploring MITRE ATT&CK

Type	Individual (hands-on)
Time Allotment	60 min
Objective	Look up techniques on ATT&CK and map behaviors to technique IDs.
Assets / Materials	Internet browser (attack.mitre.org); record answers in the CTM Virtual Lab notebook

Instructions

1. Open https://attack.mitre.org and select the Enterprise matrix.
2. For each behavior below, find the matching technique and record its ID and name.
3. Open one technique page and note one "Detection" or "Mitigation" recommendation.

Work Space

Observed Behavior	Tactic	Technique ID	Technique Name
User opened a malicious email attachment			
PowerShell ran an encoded command			
Malware created a Run registry key			
Attacker dumped passwords from LSASS memory			
Attacker used RDP to log in to the file server			
Files were encrypted and a ransom note was dropped			

One detection/mitigation recommendation you found:

Deliverable

Completed ATT&CK mapping table (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Technique IDs are correct	☐	☐
Can navigate the ATT&CK website independently	☐	☐

Activity 3.4 — IOC or IOA?

Type	Pair
Time Allotment	20 min
Objective	Differentiate indicators of compromise and indicators of attack.
Assets / Materials	None (worksheet only)

Instructions

1. Classify each item as IOC or IOA.
2. Explain why IOAs are more durable for detection.

Work Space

Indicator	IOC / IOA
SHA-256 hash of a known ransomware file	
Excel.exe starting cmd.exe	
Domain "update-microsoft-secure.example" in DNS logs	
30 failed logins followed by 1 successful login	
File named README_RESTORE_FILES.txt on the desktop	
vssadmin delete shadows /all /quiet	

Deliverable

Completed table and explanation (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
At least 5 of 6 correctly classified	☐	☐

Self-Check

A. Multiple Choice

Encircle the letter of the best answer.

1. Malware that spreads by itself across a network is a:
    a) a. Trojan
    b) b. Worm
    c) c. Rootkit
    d) d. Spyware
2. In the Cyber Kill Chain, installing persistence happens during:
    a) a. Delivery
    b) b. Exploitation
    c) c. Installation
    d) d. Reconnaissance
3. T1566 in MITRE ATT&CK refers to:
    a) a. Phishing
    b) b. Brute Force
    c) c. Data Encrypted for Impact
    d) d. Remote Services
4. Regular outbound connections at fixed intervals to the same IP most likely indicate:
    a) a. Software update
    b) b. C2 beaconing
    c) c. Backup job
    d) d. Port scan
5. Which ATT&CK tactic describes moving from one host to another?
    a) a. Discovery
    b) b. Lateral Movement
    c) c. Collection
    d) d. Execution

B. True or False

Write TRUE if the statement is correct, FALSE if it is not.

No.	Statement	Answer
1	A Trojan disguises itself as legitimate software.	
2	IOCs such as IP addresses are hard for attackers to change.	
3	Deleting shadow copies is a common ransomware behavior.	
4	MITRE ATT&CK techniques describe "how" an attacker achieves a goal.	

C. Short Answer

1. Name the seven phases of the Cyber Kill Chain in order.

2. Give two behaviors that indicate lateral movement.

Reflection

What was the most important thing you learned today, and how will you apply it in a Security Operations Center?

Trainer Sign-off

Overall Result	Trainer Remarks	Trainer Signature / Date
☐ Competent		
☐ Not Yet Competent		